

Perspective Article

Construction of antifouling fluorinated polymer brush via activators regenerated by electron transfer ATRP and thiol-epoxy click reaction

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ABSTRACT

In this work, fluorine containing polymer brushes was grafted on the functionalized stainless steel (SS) by activators regenerated by electron transfer atom transfer radical polymerization (ARGET-ATRP) and thiol-epoxy click reaction. PGMA brushes would be grafted on SS by coating the sheets with brominated polydopamine (PDA) as a reactive under-layer, and subsequently surface-initiate ARGET-ATRP of glycidyl methacrylate (GMA). Then the short-fluorinated alkyl chains were anchored on the PGMA backbones based on the click reaction between epoxy groups of PGMA and 1H, 1H, 2H, 2H-perfluorodecanethiol (PFDT), which endowed the surfaces of the polymers with low surface free energy. The surface compositions and morphologies of the functionalized SS have been characterized by attenuated total reflection-Fourier transform infrared (ATR-FTIR) spectroscopy, X-ray photoelectron spectroscopy (XPS), scanning electron microscope (SEM) and atomic force microscopy (AFM). The results of water contact angle analysis exhibited high hydrophobicity of the fluorinated polymer grafting modified SS surface. Bovine serum albumin (BSA) resistance performance of the functionalized SS was evaluated by BCA protein assay reagent, fluorescence microscope and XPS. The results exhibited that grafting fluoropolymer brushes modified SS with lower protein adsorption compared with the pristine SS surface, which might be served as promising environmentally friendly polymer antifouling coatings.

1. Introduction

Stainless steel (SS) has been widely used in the fields of biology, food and maritime industries because of its excellent mechanical strength as well as high corrosion resistance. However, it is susceptible to biofouling, which was caused by the undesired accumulation of micro- or macro- organism on the surface of the solid materials [1–3]. Biofouling is one of the most significant problems to maritime vessels, surgical implants and food container, which would cause material corrosion, economic losses, environmental pollution or health risk. Thus, surface modification of SS materials with antifouling coatings is an important way to tackle this problem. Many techniques for surface modification have been explored to suppress biofouling formation, including layer-by-layer deposition, painting, self-assembled monolayers (SAMs), chemical coupling and surface grafting reactions of the substrates [4–7]. Among these strategies, surface modification with polymer brushes has been approved to be an effective way to inhibit the accumulation of organisms on the substrates [8]. Until now, many

functional polymers with antifouling properties have been coated on the surface of varied solid materials, such as cationic polymers, zwitterionic polymers, hydrophilic or amphiphilic polymers, PEG based polymers, silicone or fluoropolymers [9–17]. Among these functional polymer brushes, fluoropolymer brushes are widely used as antifouling coatings because of its low surface energy and strongly hydrophobic character, which would prevent the adsorption of organisms [18–20]. At present, surface “grafting from” approaches based on controlled radical polymerization, including atom transfer radical polymerization (ATRP) and reversible addition-fragmentation chain transfer polymerization (RAFT) et al., have allowed better design of functional grafting polymer brush coatings [21]. Among these approaches, ATRP has been widely employed to produce various well-controlled grafted polymer brushes from solid substrates [22,23]. In comparison with traditional ATRP, ARGET-ATRP is a more attractive methodology due to ppm level of copper catalyst and atmosphere tolerance in the reaction system. It has been applied to a variety of solid substrates for the purpose of surface grafting modification [24–28].

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Recently, click reaction has been used as an effective way for building various topological polymers and functionalizing polymers, and several reactions have met the requirements of click chemistry, such as strain-promoted azide–alkyne cycloaddition (SPAAC), Diels–Alder reactions, copper (I) catalyzed azide alkyne cycloaddition (CuAAC) and Thiol-based click reaction [29–35]. Among these synthesis strategies, thiol-based click chemistry has been very successful because of water and oxygen tolerance, quantitative yields, mild conditions. Moreover, a variety of thiol-based precursors are readily available, including thiol–ene, thiol–yne, thiol–halogen, thiol–isocyanate and thiol–epoxy [36]. Thereinto, thiol–epoxy click chemistry is a very efficient and simple reaction, which play a vital role in quantitative functionalization of soft materials [37–39]. Thus, polymer architectures would be more versatile by using controlled radical polymerization combined with click reactions.

In present work, we first report the synthesis of fluorinated polymer brushes functionalized stainless steel (SS) by combination of PDA chemistry, ARGET-ATRP and thiol–epoxy click reaction. The schematic diagram for preparation of fluorinated polymer brushes grafting modified SS was given in Fig. 1. Firstly, SS@PDA was obtained by coating a homogeneous reactive PDA layer on SS surface. Subsequently, 2-bromoisobutryl bromide (BIBB) was coupled to the surface of SS@PDA to initiate polymerization of GMA by ARGET-ATRP. Finally, fluorinated alkyl chains would anchor on the functionalized SS by thiol–epoxy click reaction between the pendant epoxy groups and 1H, 1H, 2H, 2H-perfluorodecanethiol (PFDT). Thus, the fluorinated alkyl chain could impart the surfaces of SS with antifouling ability because of its low surface free energy.

2. Experimental

2.1. Materials

Stainless steel sheets (SS) (about $1 \times 1 \times 0.2 \text{ cm}^3$) were cut from commercially stainless-steel plates (304) and washed with ethanol before use. Dopamine hydrochloride (DA, 99%+) was purchased from Adamas-beta Reagent (Shanghai, China). Tris(hydroxymethyl)amino-methane (Tris, 99%) were obtained from ABCONE biotechnology Co., Ltd. (Shanghai, China). 2-bromoisobutryl bromide (BIBB, >98%) and N, N, N', N''-pentamethyldiethylene triamine (PMDETA, 99%) were purchased from TCI development Co., Ltd. (Shanghai, China). Copper (II) bromide (CuBr_2 , 99%), Ascorbic acid (AsAc, 99%), Sodium phosphate dibasic (Na_2HPO_4 , 99%), Sodium dihydrogen phosphate

(NaH_2PO_4 , 99%) and 1H, 1H, 2H, 2H-Perfluorodecanethiol (PFDT, 97%) were purchased from Aladdin Reagent Co., Ltd. (Shanghai, China). 4-Dimethylaminopyridine (DMAP, 99%) and Lithium hydroxide anhydrous (LiOH , 98%) were bought from Macklin Biochemical Co., Ltd. (Shanghai, China). Glycidyl methacrylate (GMA, Innochem Science & Technology Co., Ltd., Beijing, China) was purified by passing through a short column filled with neutral alumina to remove inhibitors. Bovine serum albumin (BSA) was received from Absin Bioscience Inc. (Shanghai, China). BCA reagents were purchased from Yeasen Biotech Co., Ltd. (Shanghai, China). Reporter Lysis 5× Buffer was obtained from Promega (Beijing) Biotech Co., Ltd. Bovine serum albumin labeled with fluorescein isothiocyanate (BSA-FITC) was got from Meilun Biotech Co., Ltd. (Dalian, China). Other reagents were analytical pure and used as received.

2.2. Preparation of SS@PDA

SS sheets were placed in a three-neck flask with a mixture of 200 mL Tris buffer solution (0.05 mol/L, pH = 8.5) and DA (0.4 g, 2.11 mmol). The reaction mixture was stirred at 60 °C for 24 h. Then SS sheets coated with PDA (SS@PDA) were prepared after repeated washing with deionized water and acetone. The products were dried in a vacuum oven at 40 °C for 12 h.

2.3. Preparation of SS@PDA-Br

SS@PDA sheets were added into a round-bottomed flask which contained with dichloromethane (150 mL), triethylamine (15 mL) and DMAP (0.225 g). Then BIBB was added dropwise into the flask under 0 °C, and the reaction was continued for another 24 h at room temperature under N_2 atmosphere. After washing with dichloromethane three times, –Br initiator functionalized SS sheets (SS@PDA-Br) were prepared. The final products were dried under vacuum at room temperature for 12 h.

2.4. Preparation of SS@PDA-g-PGMA

SS@PDA-Br sheets were placed in a dried Schlenk flask containing anisole (4 mL), GMA (2 mL, 15.1 mmol), PMDETA (15.4 mg, 0.089 mmol) and CuBr_2 (15.7 mg, 0.0089 mmol). The Schlenk flask was firstly degassed and refilled with N_2 three times, and subsequently, AsAc (15 mg, 0.085 mmol) was added to the flask quickly. Finally, the sealed Schlenk flask was degassed and backfilled with nitrogen once again, and

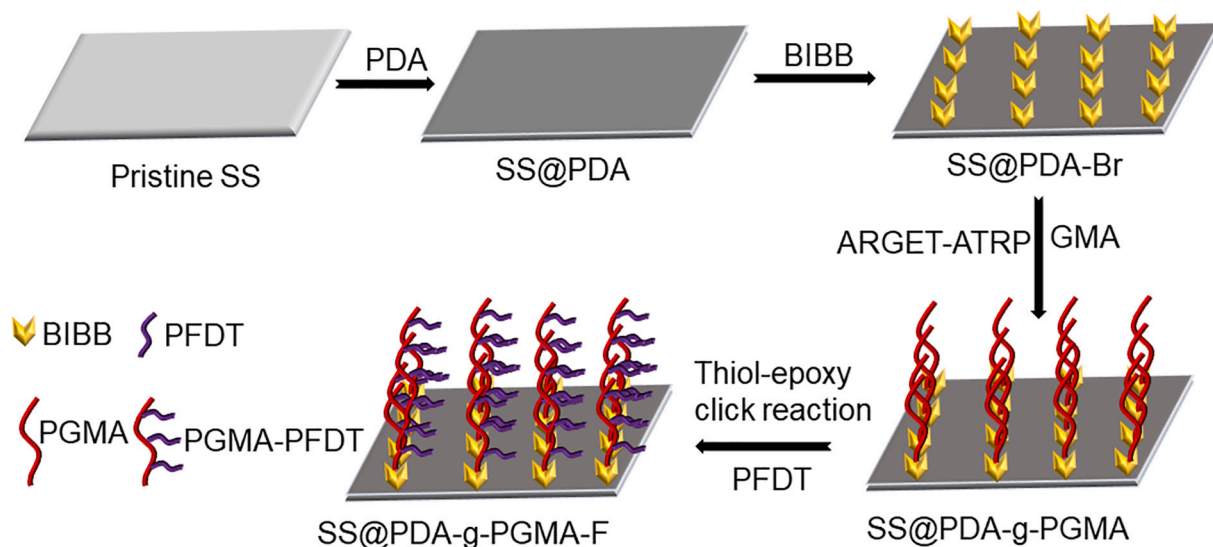


Fig. 1. Schematic diagram for surface grafting of fluorinated polymer brush on SS substrates.

the reaction was carried out at room temperature for 12 h. After washing with deionized water, ethanol and dichloromethane three times respectively, the PGMA surface modified SS sheets (SS@PDA-g-PGMA) were dried under vacuum at 40 °C for 12 h.

2.5. Preparation of SS@PDA-g-PGMA-F

A round-bottomed flask was charged with tetrahydrofuran (10 mL), LiOH (35 mg), and PFDT (86 μ L, 0.30 mmol), and then SS@PDA-g-PGMA sheets were immersed in the mixture. The reaction was carried out for 12 h at room temperature. The prepared SS@PDA-g-PGMA-F sheets were removed from the reaction mixture and washed with deionized water, acetone and dichloromethane three times respectively, and dried in a vacuum oven at 40 °C for 12 h.

2.6. Characterization

Attenuated total reflection-Fourier transform infrared (ATR-FTIR) spectra of pristine and modified SS sheets were recorded on a Nicolet-iS50 spectrometer (Thermo Fisher Scientific, USA) under total reflectance mode (range from 400 cm^{-1} to 4000 cm^{-1}). Surface compositions of the functionalized SS substrates were characterized by X-ray photoelectron spectroscopy (XPS), and all spectra were obtained by using an ESCALAB 250 Xi spectrometer (Thermo Fisher Scientific, USA) with a monochromatic Al K α X-ray source. Surface morphologies and compositions of modified SS substrates were observed by field emission scanning electron microscopy (FESEM) and energy dispersive spectroscopy (EDS) (Zeiss Merlin Compact, Germany). All samples should be gold-coated before SEM observation. Additionally, atomic force microscopy (AFM, MultiMode 8, Bruker, Germany) was used to determine the roughness and surface morphologies of the functionalized SS in a tapping mode. Water contact angles (4 μ L) of all samples were measured by contact angle analyzer (Powereach JC2000C1, Shanghai Zhongchen digital technic apparatus Co., Ltd., China).

2.7. BSA protein adsorption test

BSA was used as a model protein for evaluating the antifouling property. Pristine SS and SS@PDA-g-PGMA-F sheets were immersed into 2 mg/mL BSA/PBS solution (PBS: 0.1 M, pH = 7.4). The adsorption was incubated at 37 °C for 24 h with a shaker. After that, all samples were washed with PBS solution and deionized water three times respectively to remove weakly adsorbed BSA and PBS salt. The adsorbed protein on surface was subsequently detached in mixture of Reporter Lysis 5 $\times$ Buffer/H₂O (v : v = 1:4, 0.5 mL/2 mL) via ultrasonication at room temperature for 30 min. The concentration of the absorbed protein in Reporter Lysis 5 $\times$ Buffer was measured by using bicinchoninic acid (BCA) method [40–42]. The absorbance of protein solutions at 562 nm was measured by L55 UV–vis spectrophotometer (Shanghai Precision Scientific Instrument Co., Ltd., China), and the amount of adsorbed protein per unit area was calculated based on a calibration curve. All reported data is the average value of three parallel measurements.

BSA-FITC was dissolved in PBS (0.1 M, pH = 7.4) solution to form a protein concentration of 1 mg/mL. The bare and grafting modified SS substrates were placed in a 12-well plate and immersed in BSA-FITC solution for 24 h at 37 °C. The plate was coated with aluminum foil to provide a dark environment. The bare and modified SS substrates were then washed twice with PBS and twice with water to remove weakly adhered BSA-FITC. The adsorption of BSA-FITC on functionalized or pristine SS substrates was imaged with a Nikon Eclipse Ti–U fluorescence microscope equipped with an excitation filter of 470 nm and an emission filter of 540 nm.

Additionally, BSA resistance performance of the functionalized SS was evaluated by XPS (ESCALAB 250 Xi spectrometer), and the general procedure was as follows. Pristine and fluoropolymer grafting modified SS substrates were immersed in the BSA solution at 37 °C for 24 h in a

shaker. After washing successively with PBS and deionized water, the samples were dried under vacuum. The compositions of adsorbed protein on samples were analyzed by XPS measurements.

3. Results and discussion

3.1. Surface structures and morphologies

ATR-FTIR was preliminarily used to characterize chemical structures of the modified SS. It can be seen in Fig. 2b, new peaks of stretching vibration of C–N (1280 cm^{-1}), skeleton vibration of benzene ring (1602 cm^{-1} , 1503 cm^{-1}) and stretching vibration of N–H and O–H (around 3300 cm^{-1}) were detected in spectrum of SS@PDA (b), implying that PDA-coated SS sheets were prepared successfully. For SS@PDA-Br, new absorptions at 2920 cm^{-1} and 2851 cm^{-1} (enlarged spectrum of c) corresponding to C–H asymmetric and symmetric stretching vibration of methyl group in BIBB were observed, indicating that BIBB was immobilized on SS@PDA surface. Compared to SS@PDA-Br, new peaks at $\nu_{\text{C=O}}$ = 1725 cm^{-1} , $\nu_{\text{as C-O-C}}$ = 1256 cm^{-1} , $\nu_{\text{s C-O-C}}$ = 1131 cm^{-1} , which were attributed to the chemical groups of grafted PGMA on SS@PDA-g-PGMA (spectrum d) were observed. Moreover, characteristic peak at 908 cm^{-1} corresponding to symmetric stretching of epoxy group was also appeared in spectrum d. It demonstrated that PGMA has been grafted onto modified SS substrates successfully. After thiol-epoxy click reaction, the characteristic peak of epoxy group at 908 cm^{-1} was disappeared, while new signals located at 1199 cm^{-1} and 1147 cm^{-1} of C–F stretching vibration were appeared in spectrum (e), illustrating that PFDT was introduced onto SS@PDA-g-PGMA surface. Thus, SS@PDA-g-PGMA-F was obtained. The results of ATR-FTIR confirmed the successful preparation of fluorinated polymer brush functionalized SS substrates.

Surface composition of modified SS sheets was also determined by XPS to demonstrate the successful surface modification of SS, and the results were given in Figs. 3 and 4. The signals of C 1s and O 1s could be detected on pristine SS substrate, and the peak at 532.8 eV of O 1s might be ascribed to Fe₂O₃ layer (Figs. 3(a), and 4(a, O 1s)). For SS@PDA, N 1s peak at 400.2 eV was appeared in Fig. 3(b) and Fig. 4 (b, N 1s). Additionally, the characteristic signal at 710.5 eV of Fe 2p are no longer discernible, indicating SS has been coated by PDA layer. After –Br initiator was anchored on the surface of functionalized SS, signal at 70.7 eV that attributed to Br 3d was observed in Fig. 3(c) and Fig. 4(c, Br 3d). Additionally, C 1s spectrum of SS@PDA-Br have been curve fitted into three peaks with binding energies at 288.8 eV, 286.5 eV and 284.8 eV for the respective C=O, C–O/C–Br and C–C/C–H. The O 1s peaks of the fitted curves at 533.7 eV and 532.3 eV which were assigned to C–O and C=O bonds have also been detected in XPS spectrum of SS@PDA-Br (Fig. 4(c, O 1s)). After graft polymerization on SS@PDA-Br substrates, the intensity of O 1s peak was increased distinctively (Figs. 3(d), and 4 (d, O 1s)) due to the grafted PGMA containing oxygen-rich species. Moreover, the C 1s peak (Fig. 4(d, C 1s)) was split into three peaks at 288.8 eV, 286.5 eV, 284.8 eV, which are attributable to C=O, C–O and C–C/C–H, respectively. For SS@PDA-g-PGMA-F, a strong peak at 689.5 eV of F 1s was emerged (Fig. 3(e), and Fig. 4(e, F 1s)) and binding energy of C 1s at 292.4 eV and 294.6 eV which resulted from C–F and C–S bonds respectively were also detected, implying the modified SS has been successfully covered by fluorinated polymer brush.

The results of element percentage on the surface of modified SS based on XPS analysis were listed in Table. 1. The element content of N 1s for SS@PDA was increased to 8.48% which was significantly higher than that pristine SS, confirming the successful coating of PDA. The element of Br 3d was detected on surface of SS@PDA-Br, and its percentage was increased to 2.45%. However, the Br 3d content was reduced to 0.16% after grafting of PGMA on SS@PDA-Br. What's more, the surface element percentage of O 1s was enhanced to 27.07% for SS@PDA-g-PGMA. After thiol-epoxy click reaction between the grafted PGMA and PFDT, element percentage of F 1s was increased sharply with

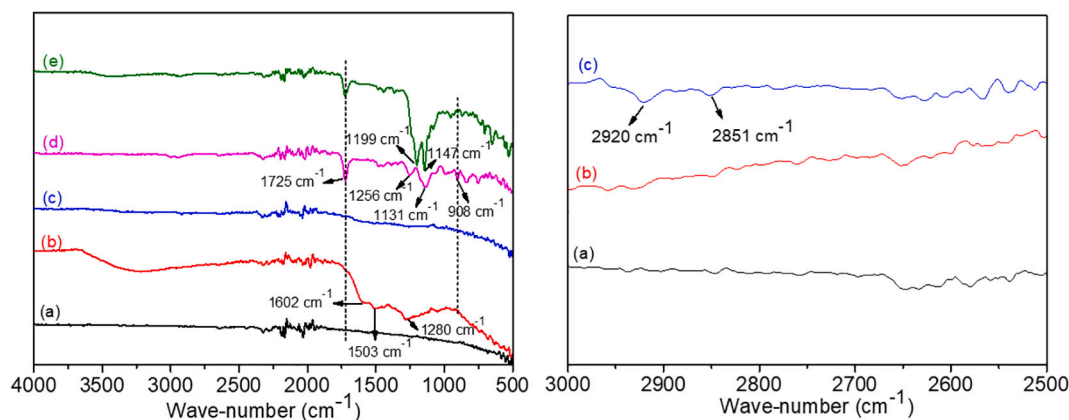


Fig. 2. ATR-FTIR spectra of pristine SS (a), SS@PDA (b), SS@PDA-Br (c), SS@PDA-g-PGMA (d) and SS@PDA-g-PGMA-F (e); The right graph is the enlarged ATR-FTIR spectra.

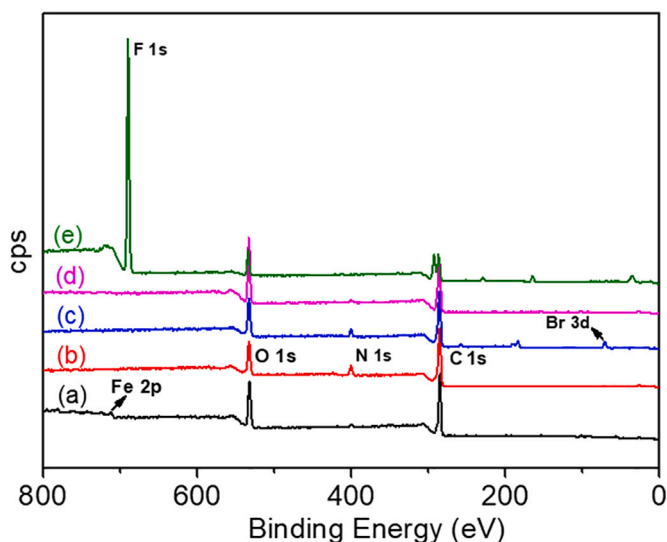


Fig. 3. XPS spectra of pristine SS (a), SS@PDA (b), SS@PDA-Br (c), SS@PDA-g-PGMA (d) and SS@PDA-g-PGMA-F (e).

a value of 50.65%. All XPS results proved that each step reaction for SS surface modification was successful. Thus, the modified SS substrates have been covered by fluorinated polymer brush.

SEM images for surface morphologies of modified SS were shown in Fig. 5. A relatively flat surface could be observed on surface of bare SS (image a). After coating with PDA, a lot of clusters were presented on the surface of SS@PDA (image b). While number of clusters was decreased on the surface of SS@PDA-Br, and larger convex structure was formed after BIBB anchored on SS@PDA surface (image c). Instead of clusters, a rather denser and rougher surface was observed on surface of SS@PDA-g-PGMA, which might be attributed to the aggregate PGMA chains covered on the modified SS (image d). The introduction of PFDT chains on modified SS substrate resulted in forming of a smoother surface (image e) compared to that of SS@PDA-g-PGMA. Surface chemical compositions of SS@PDA-g-PGMA-F were also investigated by EDS analysis (images f-k), and elements such as C, N, O, Br, F were all detected. It can be seen clearly the intensity of F in image k was very high, which is in consistent with the results of XPS.

AFM was also employed to exhibit the surface morphologies and roughness of modified SS, and the images were given as Fig. 6. The surface of pristine SS exhibited a root mean square roughness (RMS) of 24.0 nm for a $2\ \mu\text{m} \times 2\ \mu\text{m}$ scan (image a). After PDA coated on the surface of SS substrates, RMS value of SS@PDA (b) was increased to 163

nm. The RMS value of SS@PDA-Br was 69.1 nm and increased to 135 nm after grafting PGMA polymer brushes, indicating a rougher surface for SS@PDA-g-PGMA. However, the roughness (R_{RMS} 129 nm) of SS@PDA-g-PGMA-F (image e) was slightly decreased after PFDT was grafted onto SS@PDA-g-PGMA surface. The obtained results from AFM analysis were in agreement with that of aforementioned SEM observations, and it further demonstrated that the surface of SS substrate has been tailored by fluoropolymer brushes.

Contact angle measurement was carried out to determine surface hydrophilicity/hydrophobicity of modified SS sheets, and the results were shown in Fig. 7. The contact angle of pristine SS was 73° (A) and decreased to 10° (B) after coating with hydrophilic PDA. While the contact angle was elevated to 104° (C) for SS@PDA-Br, which might be originated from the hydrophobic alkyl groups in BIBB. For SS@PDA-g-PGMA, the contact angle was about 74° (D) which was lower than SS@PDA-Br, owing to the pendant carbonyl and epoxy groups located at the grafted PGMA chains. However, the introduced hydrophobic PFDT chains increased the contact angle of SS@PDA-g-PGMA-F samples to 131° (E). The results of contact angle test implied that the changes of surface hydrophilicity/hydrophobicity for modified SS substrates much depended on the surface structure, and the grafting fluorinated polymer showed highly hydrophobic.

3.2. Antifouling ability

BSA was served as the protein model to explore the antifouling ability of the grafted fluoropolymer functionalized SS. The protein resistance performance was first determined by BCA protein assay reagent, and the value given here was the average of three parallel measurements. The assay results of BSA resistance performance for pristine and modified SS were displayed in Fig. 8. It showed that the adhered amount of BSA proteins on SS@PDA-g-PGMA-F dropped nearly 27% compared with that of pristine SS. The reduced BSA protein adsorption might be attributed to the highly hydrophobic property of the grafted short fluorinated alkyl chain.

The fluorescence images of BSA-FITC adsorbed on pristine SS and SS@PDA-g-PGMA-F were shown in Fig. 9. After exposed to BSA-FITC aqueous solution for 24 h, the pristine SS was heavily fouled by BSA protein and there was a lot of BSA-FITC adsorbed on pristine SS (Fig. 9 a-c). In contrast, little discrete protein that adsorbed on the surface of SS@PDA-g-PGMA-F was observed (Fig. 9 d-f).

The protein adsorption resistance of the grafting fluoropolymer brush was also evaluated based on XPS analysis. The XPS spectra of pristine and modified SS samples after 24 h BSA adsorption were shown in Fig. 10. The intensity of N 1s peak was significantly increased compared to the spectrum of pristine SS before BSA adsorption (Fig. 10a and Fig. 3a). The $[N]/[C]$ ratio could be used to represent the relative

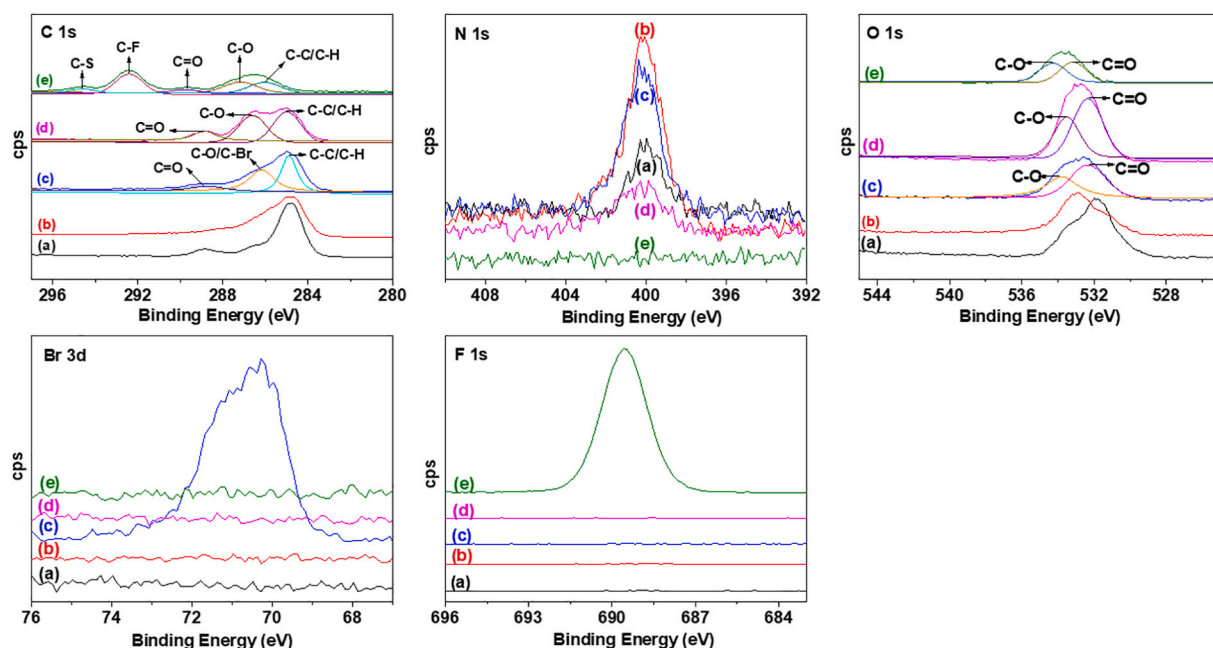


Fig. 4. High resolution XPS scan of C 1s, N 1s, O 1s, Br 3d and F 1s of pristine SS (a), SS@PDA (b), SS@PDA-Br (c), SS@PDA-g-PGMA (d) and SS@PDA-g-PGMA-F (e).

Table 1

Element contents of SS, SS@PDA, SS@PDA-Br, SS@PDA-g-PGMA and SS@PDA-g-PGMA-F determined by XPS.

Samples	C%	N%	O%	Br%	F%
SS	71.94	2.93	23.76	0.17	1.19
SS@PDA	73.29	8.48	17.41	0.14	0.68
SS@PDA-Br	70.73	6.51	19.08	2.45	1.23
SS@PDA-g-PGMA	68.65	3.08	27.07	0.16	1.03
SS@PDA-g-PGMA-F	40.61	0.86	7.7	0.18	50.65

absorbed amount of BSA on surfaces of SS substrate after exposure to BSA solution [43]. The [N]/[C] ratio of bare SS increased from 0.041 (according to Table 1) to 0.145 after immersion in BSA solution for 24 h, indicating that pristine SS has been fouled by BSA protein. However, [N]/[C] ratio of SS@PDA-PGMA-F was 0.021 and 0.072 before and after exposure to BSA solution. The increment of [N]/[C] ratio for SS@PDA-PGMA-F was lower than that of bare SS. Moreover, the characteristic signal of amide ($O=CNH$) at 288.5 eV in the fitted curve of C 1s spectrum was clearly observed on the pristine SS surface after BSA adsorption, but the signals of amide was not obviously discerned in XPS spectrum of

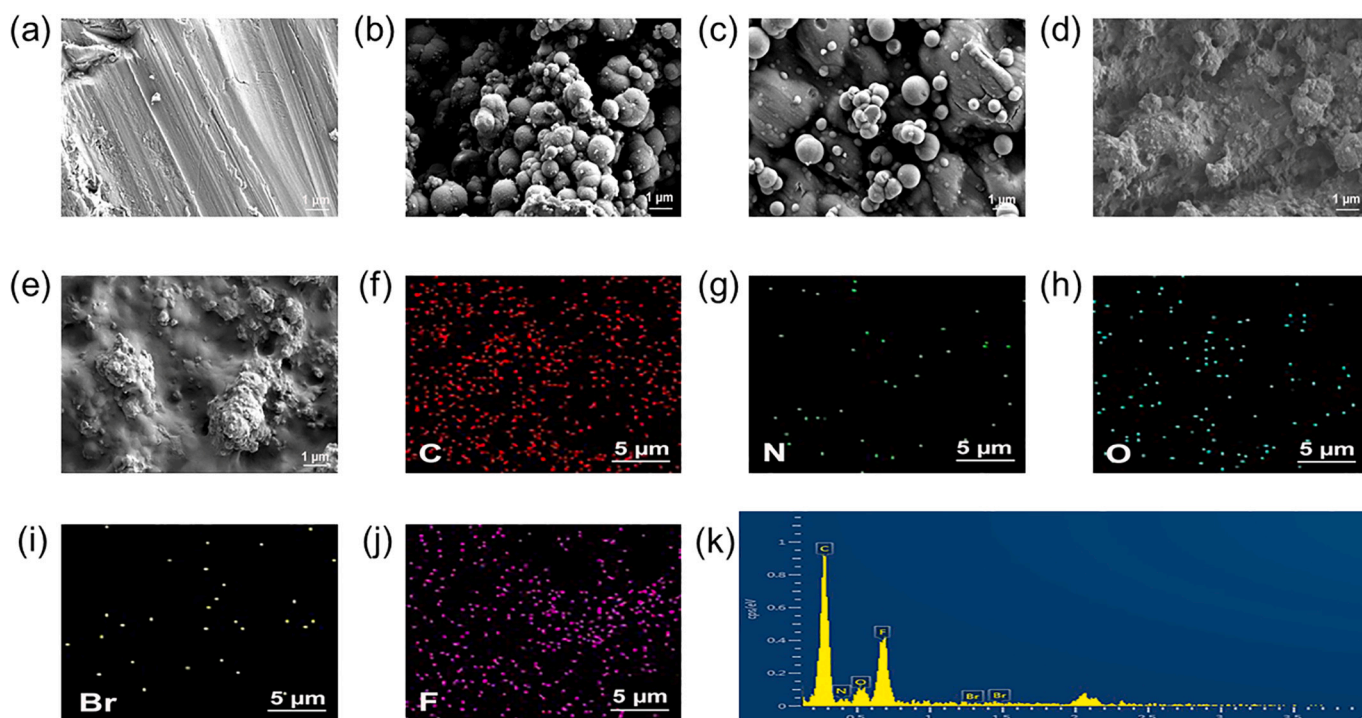


Fig. 5. SEM images for surface morphologies from pristine SS (a), SS@PDA (b), SS@PDA-Br (c), SS@PDA-g-PGMA (d) and SS@PDA-g-PGMA-F (e); f-k are EDS element mapping images and spectra for SS@PDA-g-PGMA-F.

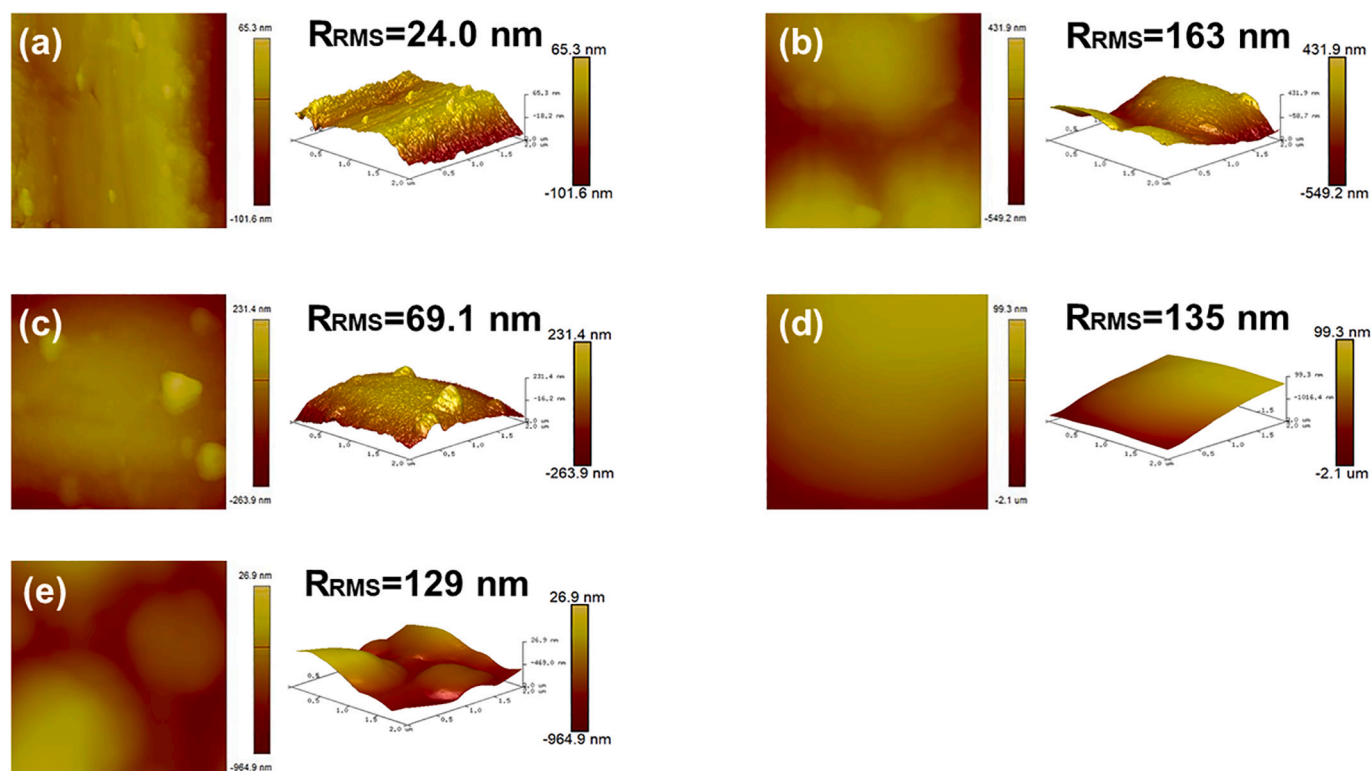


Fig. 6. AFM surface images and roughness of pristine SS (a), SS@PDA (b), SS@PDA-Br (c), SS@PDA-g-PGMA (d) and SS@PDA-g-PGMA-F (e); The dimension of each scan image is $2 \times 2 \mu\text{m}^2$.

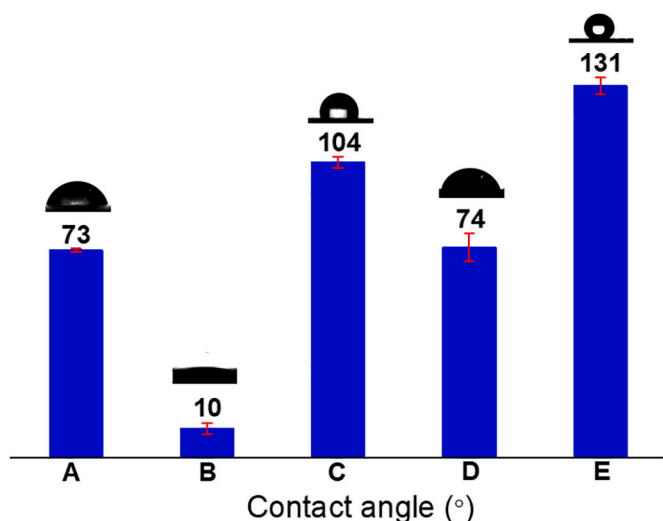


Fig. 7. Water contact angles of pristine SS (A), SS@PDA (B), SS@PDA-Br (C), SS@PDA-g-PGMA (D) and SS@PDA-g-PGMA-F (E); Inset pictures are images of water droplets on different substrates.

SS@PDA-g-PGMA-F after BSA absorption. All the mentioned above results confirmed that the grafted fluoropolymer brushes have a decisive effect on decreasing protein adsorption of the modified SS.

4. Conclusions

Structure controllable fluorine containing polymer brush was grafted onto SS surface successfully by combination ARGET-ATRP with thiol-epoxy click chemistry for the first time. The water contact angle value of fluorinated polymer brushes grafting modified SS was increased to

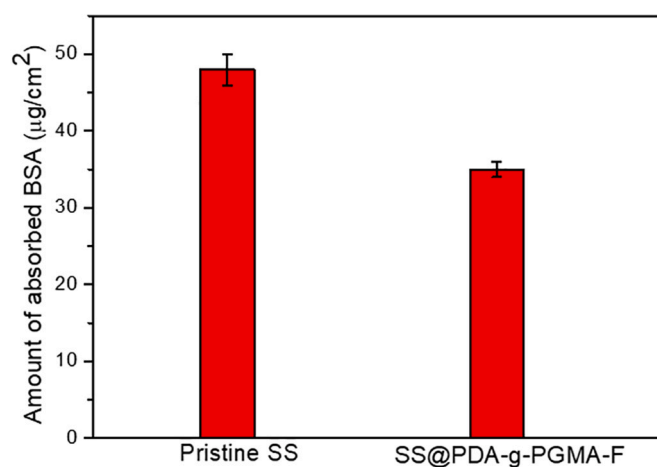


Fig. 8. The absorbed amount of BSA proteins on surfaces of pristine SS and SS@PDA-g-PGMA-F.

131° , which exhibited high hydrophobicity. The fluoropolymer grafting modified surface showed $35 \mu\text{g}/\text{cm}^2$ of BSA adsorption, which was 27% lower than that of pristine SS ($48 \mu\text{g}/\text{cm}^2$). All the antifouling tests confirmed that SS@PDA-g-PGMA-F showed higher resistance against BSA in comparison with that of pristine SS, demonstrating the good antifouling behavior. The combination of these two step processes described in this work provides a simple method for fabricating hydrophobic polymer brushes with antifouling ability on stainless steel surfaces, and it represents an effective and green technology to fabricate other functional polymer brush coatings on varied solid material surfaces for advanced applications.

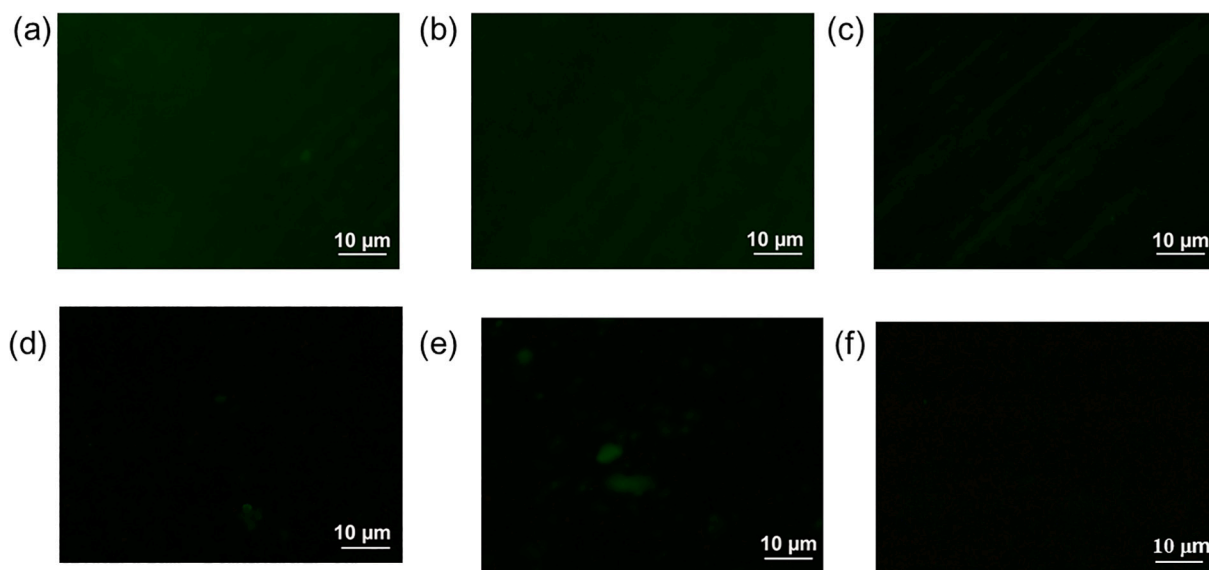


Fig. 9. Fluorescence micrographs of BSA-FITC adhered to pristine SS (a-c) and SS@PDA-g-PGMA-F (d-f).

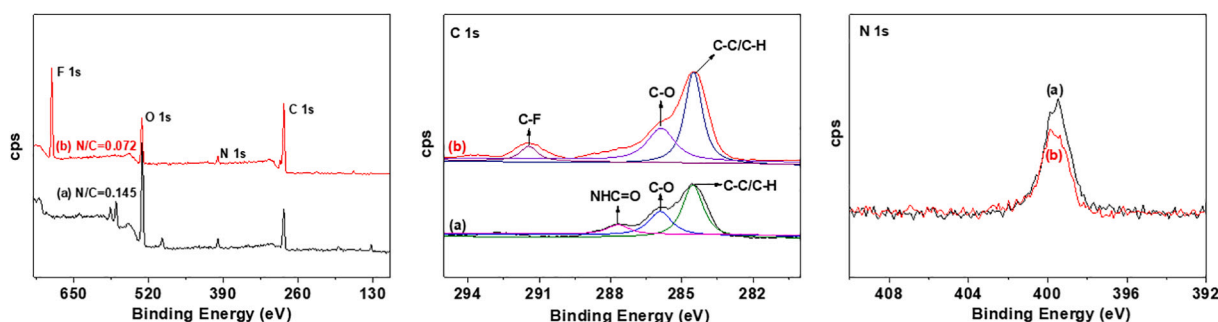


Fig. 10. XPS spectra and high resolution of C 1s, N 1s of pristine SS (a) and SS@PDA-g-PGMA-F (b) after exposure to BSA solutions for 24 h (washed by PBS after protein adsorption).

Data availability

The authors confirm that the data supporting the findings of this study are available within the article.

Declaration of Competing Interest

The authors declare no conflict of interest.

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References

- [1] K.G. Neoh, E.T. Kang, Combating bacterial colonization on metals via polymer coatings: relevance to marine and medical applications, *ACS Appl. Mater. Interfaces* 3 (2011) 2808–2819, <https://doi.org/10.1021/am200646t>.
- [2] Q. Sun, H. Li, C. Xian, Y. Yang, Y. Song, P. Cong, Mimetic marine antifouling films based on fluorine-containing polymethacrylates, *Appl. Surf. Sci.* 344 (2015) 17–26, <https://doi.org/10.1016/j.apsusc.2015.03.101>.
- [3] J. Landoulsi, K.E. Cooksey, V. Dupres, Review-interactions between diatoms and stainless steel: focus on biofouling and biocorrosion, *Biofouling* 27 (2011) 1105–1124, <https://doi.org/10.1080/08927014.2011.629043>.
- [4] L.Q. Xu, D. Pranantyo, J.B. Liu, K.-G. Neoh, E.-T. Kang, Y.X. Ng, S. Lay-Ming Teo, G.D. Fu, Layer-by-layer deposition of antifouling coatings on stainless steel via catechol-amine reaction, *RSC Adv.* 4 (2014) 32335–32344, <https://doi.org/10.1039/C4RA04336G>.
- [5] F. Wang, H. Zhang, B. Yu, S. Wang, Y. Shen, H. Cong, Review of the research on anti-protein fouling coatings materials, *Prog. Org. Coat.* 147 (2020) 105860, <https://doi.org/10.1016/j.porgcoat.2020.105860>.
- [6] P. Cao, K. Liu, X. Liu, W. Sun, D. Wu, C. Yuan, X. Bai, C. Zhang, Antibacterial properties of Magainin II peptide onto 304 stainless steel surfaces: a comparison study of two dopamine modification methods, *Colloids Surf. B: Biointerfaces* 194 (2020) 111198, <https://doi.org/10.1016/j.colsurfb.2020.111198>.
- [7] X. Fan, L. Lin, J.L. Dalsin, P.B. Messersmith, Biomimetic anchor for surface-initiated polymerization from metal substrates, *J. Am. Chem. Soc.* 127 (2005) 15843–15847, <https://doi.org/10.1021/ja0532638>.
- [8] W.J. Yang, K.-G. Neoh, E.-T. Kang, S.L.-M. Teo, D. Rittschof, Polymer brush coatings for combating marine biofouling, *Prog. Polym. Sci.* 39 (2014) 1017–1042, <https://doi.org/10.1016/j.progpolymsci.2014.02.002>.
- [9] B. Zhang, Q. Yan, S. Yuan, X. Zhuang, F. Zhang, Enhanced antifouling and anticorrosion properties of stainless steel by biomimetic anchoring PEGDMA-cross-linking Polycationic brushes, *Ind. Eng. Chem. Res.* 58 (2019) 7107–7119, <https://doi.org/10.1021/acs.iecr.8b05599>.
- [10] D. Pranantyo, L.Q. Xu, K.-G. Neoh, E.-T. Kang, Y.X. Ng, S.L.-M. Teo, Tea stains-inspired initiator primer for surface grafting of antifouling and antimicrobial polymer brush coatings, *Biomacromolecules* 16 (2015) 723–732, <https://doi.org/10.1021/bm501623c>.
- [11] Z. Nadizadeh, H. Mahdavi, Grafting of zwitterion polymer on polyamide nanofiltration membranes via surface-initiated RAFT polymerization with improved antifouling properties as a new strategy, *Sep. Purif. Technol.* 254 (2021) 117605, <https://doi.org/10.1016/j.seppur.2020.117605>.
- [12] H.-X. Wu, X.-H. Zhang, L. Huang, L.-F. Ma, C.-J. Liu, Diblock polymer brush (PHEAA-b-PFMA): microphase separation behavior and anti-protein adsorption performance, *Langmuir* 34 (2018) 11101–11109, <https://doi.org/10.1021/acs.langmuir.8b02584>.
- [13] L.Q. Xu, D. Pranantyo, Y.X. Ng, S.L.-M. Teo, K.-G. Neoh, E.-T. Kang, G.D. Fu, Antifouling coatings of catecholamine copolymers on stainless steel, *Ind. Eng. Chem. Res.* 54 (2015) 5959–5967, <https://doi.org/10.1021/acs.iecr.5b00171>.

- [14] J. Wang, H. Sun, X. Gao, C. Gao, Enhancing antibiofouling performance of Polysulfone (PSF) membrane by photo-grafting of capsaicin derivative and acrylic acid, *Appl. Surf. Sci.* 317 (2014) 210–219, <https://doi.org/10.1016/j.apsusc.2014.08.102>.
- [15] B. Xu, Y. Liu, X. Sun, J. Hu, P. Shi, X. Huang, Semifluorinated synergistic nonfouling/fouling-release surface, *ACS Appl. Mater. Interfaces* 9 (2017) 16517–16523, <https://doi.org/10.1021/acsami.7b03258>.
- [16] J. Li, G. Wang, C. Ding, H. Jiang, P. Wang, Synthesis and evaluation of polystyrene-polybutadiene-polystyrene-dodecafluoroheptyl methacrylate/polystyrene-polybutadiene-polystyrene hybrid antifouling coating, *J. Colloid Interface Sci.* 434 (2014) 71–76, <https://doi.org/10.1016/j.jcis.2014.07.043>.
- [17] Z. Lu, Z. Chen, Y. Guo, Y. Ju, Y. Liu, R. Feng, C. Xiong, C.K. Ober, L. Dong, Flexible hydrophobic antifouling coating with oriented Nanotopography and nonleaking capsaicin, *ACS Appl. Mater. Interfaces* 10 (2018) 9718–9726, <https://doi.org/10.1021/acsami.7b19436>.
- [18] N.S. Bhairamadgi, S.P. Pujari, C.J.M. van Rijn, H. Zuithof, Adhesion and friction properties of Fluoropolymer brushes: on the Tribological inertness of fluorine, *Langmuir* 30 (2014) 12532–12540, <https://doi.org/10.1021/la501802b>.
- [19] Z. Wang, H. Zuithof, Antifouling properties of Fluoropolymer brushes toward organic polymers: the influence of composition, thickness, brush architecture, and annealing, *Langmuir* 32 (2016) 6571–6581, <https://doi.org/10.1021/acs.langmuir.6b00695>.
- [20] Y. Wang, Y. Liu, M. Liu, W. Qian, D. Zhou, T. Liu, G. Luo, M. Xing, Short fluorocarbon chains containing hydrophobic nanofibrous membranes with improved hemocompatibility, anticoagulation and anti-fouling performance, *Colloids Surf. B: Biointerfaces* 180 (2019) 49–57, <https://doi.org/10.1016/j.colsurfb.2019.01.025>.
- [21] R. Barbey, L. Lavanant, D. Paripovic, N. Schüwer, C. Sugnaux, S. Tugulu, H.-A. Klok, Polymer brushes via surface-initiated controlled radical polymerization: synthesis, characterization, properties, and applications, *Chem. Rev.* 109 (2009) 5437–5527, <https://doi.org/10.1021/cr900045a>.
- [22] W. Zhao, Q. Ye, H. Hu, X. Wang, F. Zhou, Grafting zwitterionic polymer brushes via electrochemical surface-initiated atomic-transfer radical polymerization for anti-fouling applications, *J. Mater. Chem. B* (2014) 5352–5357, <https://doi.org/10.1039/C4TB00816B>.
- [23] B.R. Coad, Y. Lu, L. Meagher, A substrate-independent method for surface grafting polymer layers by atom transfer radical polymerization: reduction of protein adsorption, *Acta Biomater.* 8 (2012) 608–618, <https://doi.org/10.1016/j.actbio.2011.10.006>.
- [24] B. Wang, Z. Wang, F. Jiang, H. Fang, Z. Wang, Synthesis and characterization of MWCNT-graft-polyisoprene via ARGET ATRP, *RSC Adv.* 4 (2014) 26468–26475, <https://doi.org/10.1039/C4RA02986K>.
- [25] S. Wang, H. Meng, Y. Li, D. Sun, Y. Zhan, X. Ge, L. Chen, Polymer brushes grafted from graphene via bioinspired polydopamine chemistry and activators regenerated by electron transfer atom transfer radical polymerization, *J. Polym. Sci. Part A: Polym. Chem.* 57 (2019) 689–698, <https://doi.org/10.1002/pola.29310>.
- [26] Y. Song, G. Ye, Y. Lu, J. Chen, J. Wang, K. Matyjaszewski, Surface-initiated ARGET ATRP of poly(Glycidyl methacrylate) from carbon nanotubes via bioinspired catechol chemistry for efficient adsorption of uranium ions, *ACS Macro Lett.* 5 (2016) 382–386, <https://doi.org/10.1021/acsmacrolett.6b00099>.
- [27] T.B. Mai, T.N. Tran, M. Rafiqul Islam, J.M. Park, K.T. Lim, Covalent functionalization of silica nanoparticles with poly(N-isopropylacrylamide) employing thiol-ene chemistry and activator regenerated by electron transfer ATRP protocol, *J. Mater. Sci.* 49 (2014) 1519–1526, <https://doi.org/10.1007/s10853-013-7833-4>.
- [28] S. Wang, J. Song, Y. Li, X. Zhao, L. Chen, G. Li, L. Wang, Z. Jia, X. Ge, Grafting antibacterial polymer brushes from titanium surface via polydopamine chemistry and activators regenerated by electron transfer ATRP, *React. Funct. Polym.* 140 (2019) 48–55, <https://doi.org/10.1016/j.reactfunctpolym.2019.04.002>.
- [29] M. Meldal, F. Diness, Recent fascinating aspects of the CuAAC click reaction, *Trends Chem.* 2 (2020) 569–584, <https://doi.org/10.1016/j.trechm.2020.03.007>.
- [30] M.D. Hossain, Z. Jia, M.J. Monteiro, Complex polymer topologies built from tailored multifunctional cyclic polymers, *Macromolecules* 47 (2014) 4955–4970, <https://doi.org/10.1021/ma501049n>.
- [31] X. Liu, Y. Wu, M. Zhang, K. Zhang, Efficient polymer dimerization method based on self-accelerating click reaction, *RSC Adv.* 10 (2020) 6794–6800, <https://doi.org/10.1039/C9RA09919K>.
- [32] P. Sun, J. Chen, J.A. Liu, K. Zhang, Self-accelerating click reaction for cyclic polymer, *Macromolecules* 50 (2017) 1463–1472, <https://doi.org/10.1021/acs.macromol.6b02614>.
- [33] G.S. Dajana, S.C. Marija, Diels-Alder “click” reactions, *Curr. Org. Chem.* 20 (2016) 2211–2221, <https://doi.org/10.2174/1385272820666160215235852>.
- [34] M.A. Tasdelen, Diels-Alder “click” reactions: recent applications in polymer and material, *Polym. Chem.* 2 (2011) 2133–2145, <https://doi.org/10.1039/C1PY00041A>.
- [35] A.B. Lowe, M.A. Harvison, Thiol-based click chemistries in polymer synthesis and modification, *Aust. J. Chem.* 63 (2010) 1251–1266, <https://doi.org/10.1071/CH10214>.
- [36] W. Zhang, K. Chen, G. Chen, in: S. Chandrasekaran (Ed.), Thiol-Based “Click” Chemistry for Macromolecular Architecture Design, Click Reactions in Organic Synthesis, 2016, <https://doi.org/10.1002/9783527694174.ch8>.
- [37] I. Gadwal, A. Khan, Multiply functionalized dendrimers: protective-group-free synthesis through sequential thiol-epoxy ‘click’ chemistry and esterification reaction, *RSC Adv.* 5 (2015) 43961–43964, <https://doi.org/10.1039/C5RA05820A>.
- [38] S. De, A. Khan, Efficient synthesis of multifunctional polymers via thiol-epoxy “click” chemistry, *Chem. Commun.* 48 (2012) 3130–3132, <https://doi.org/10.1039/C2CC30434A>.
- [39] A. Brändle, A. Khan, Thiol-epoxy ‘click’ polymerization: efficient construction of reactive and functional polymers, *Polym. Chem.* 3 (2012) 3224–3227, <https://doi.org/10.1039/C2PY20591B>.
- [40] P.K. Smith, R.I. Krohn, G.T. Hermanson, A.K. Mallia, F.H. Gartner, M. D. Provenzano, E.K. Fujimoto, N.M. Goeke, B.J. Olson, D.C. Klenk, Measurement of protein using Bicinchoninic acid, *Anal. Biochem.* 150 (1985) 76–85, [https://doi.org/10.1016/0003-2697\(85\)90442-7](https://doi.org/10.1016/0003-2697(85)90442-7).
- [41] P. Khramtsov, T. Kalashnikova, M. Bochkova, M. Kropaneva, V. Timganova, S. Zamorina, M. Rayev, Measuring the concentration of protein nanoparticles synthesized by desolvation method: comparison of Bradford assay, BCA assay, hydrolysis/UV spectroscopy and gravimetric analysis, *Int. J. Pharm.* 599 (2021) 120422, <https://doi.org/10.1016/j.ijpharm.2021.120422>.
- [42] L. Chen, Z. Yu, Y. Lee, X. Wang, B. Zhao, Y.M. Jung, Quantitative evaluation of proteins with bicinchoninic acid (BCA): resonance Raman and surface-enhanced resonance Raman scattering-based methods, *Analyst* 137 (2012) 5834–5838, <https://doi.org/10.1039/C2AN35715A>.
- [43] S. Yuan, D. Wan, B. Liang, S.O. Pehkonen, Y.P. Ting, K.G. Neoh, E.T. Kang, Lysozyme-coupled poly(poly(ethylene glycol) methacrylate)–stainless steel hybrids and their antifouling and antibacterial surfaces, *Langmuir* 27 (2011) 2761–2774, <https://doi.org/10.1021/la104442f>.